

Calculating Cosine Similarity with Matrix Decomposition (matrix multiplication with normalized columns)

Asked 7 years, 3 months ago Modified 7 years, 3 months ago Viewed 3k times



2



To calculate the **column cosine similarity** of $\mathbf{R} \in \mathbb{R}^{m \times n}$, $\mathbf{R}$ is normalized by Norm2 of their columns, then the cosine similarity is calculated as

$$\text{cosine similarity} = \bar{\mathbf{R}}^\top \bar{\mathbf{R}}.$$

where $\bar{\mathbf{R}}$ is the normalized $\mathbf{R}$,

If I have $\mathbf{U} \in \mathbb{R}^{m \times l}$ and $\mathbf{P} \in \mathbb{R}^{n \times l}$ defined as

$$\mathbf{R} = \mathbf{U}\mathbf{P}^\top$$

where l is the number of latent values. To calculate the similarity, multiply them and use the above equation.

But if $m \gg n$ and $m, n \gg l$, it's very inefficient. So I tried the flowing expansion:

$$\mathbf{R}^\top \mathbf{R} = \mathbf{P}\mathbf{U}^\top \mathbf{U}\mathbf{P}^\top$$

$$\mathbf{U}^\top \mathbf{U} \in \mathbb{R}^{l \times l} = \mathbf{L}\mathbf{L}^* \text{ by cholesky decomposition}$$

$$\mathbf{R}^\top \mathbf{R} = \mathbf{P}\mathbf{L}\mathbf{L}^\top \mathbf{P}^\top := \mathbf{S}^\top \mathbf{S}.$$

where $\mathbf{S} = (\mathbf{P}\mathbf{L})^\top \in \mathbb{R}^{l \times n}$.

Now, analogous to the above, $\mathbf{S}$ is normalized by Norm2 their columns, then the similarity would be

$$\text{cosine similarity} = \bar{\mathbf{S}}^\top \bar{\mathbf{S}}.$$

where $\bar{\mathbf{S}}$ is the normalized $\mathbf{S}$.

My question is **two results are the same?**

$$\bar{\mathbf{R}}^\top \bar{\mathbf{R}} \approx \bar{\mathbf{S}}^\top \bar{\mathbf{S}}$$

I think because the normalizing columns are different, it will be different. But I tried toy example(using Python/numpy), they are the same. Is it correct?

```

import numpy as np

u = 10000 # users
p = 1000 # products
l = 100 # latents

U = np.random.randn(u, l) # user feature

P = np.random.randn(p, l) # product feature
R = np.dot(U, P.T) # user-product interaction

Rbar = R / np.linalg.norm(R, 2, 0)

direct_similarity = np.dot(Rbar.T, Rbar)

L = np.linalg.cholesky(np.dot(U.T, U))
PL = np.dot(P, L)

PLbar = (PL.T / np.linalg.norm(PL, 2, 1)).T

indirect_similarity = np.dot(PLbar, PLbar.T)

np.allclose(direct_similarity, indirect_similarity)
>>> True

```

matrix

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asked Jul 6, 2015 at 4:40



emeth

23 1 7



I wrote the same question in math.stackexchange.com/questions/1349971/... – emeth Jul 6, 2015 at 4:41

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1 Answer



The normalizing factors are the same.

2

Let us take two matrices $R \in \mathbb{R}^{m \times n}$, $S \in \mathbb{R}^{l \times n}$ such that $R^T R = S^T S$. Now, $\bar{R}$ can be obtained by multiplying R with an appropriate diagonal matrix D_R from the right:

$\bar{R} = R D_R$, where

$$D_R^2 = \sum_{k=1}^n I_n^{(k)} R^T R I_n^{(k)},$$



+100



where $I_n^{(k)}$ is an $n \times n$ matrix with zeros everywhere, except for the element at (k, k) , where it has a 1. Since $R^T R = S^T S$, we have $D_R^2 = D_S^2$, thus (being diagonal matrices with non-negative elements) $D_R = D_S$.

Hence

HENCE,

$$\bar{R}^T \bar{R} = (RD_R)^T (RD_R) = D_R R^T R D_R = D_S S^T S D_S = \bar{S}^T \bar{S}.$$

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edited Jul 13, 2015 at 20:33

answered Jul 13, 2015 at 20:18

sandris



526 3 7

▲ thanks. I am not familiar with this area. could you recommend any reference or article in online?
▢ – emeth Jul 14, 2015 at 7:01

▲ Do you mean to explain the answer better, or matrix algebra, decompositions, etc. in general?
▢ – sandris Jul 14, 2015 at 7:29

▲ All I need is matrix algebra .:) – emeth Jul 14, 2015 at 9:32
▢

▲ Well, yes, for this answer all we use is matrix multiplication, addition and transposes. Judging from your question and code you already handle these confidently. A web search brings up a few online tutorials on matrices, but I haven't used any of them, so cannot recommend one. – sandris Jul 14, 2015 at 11:05
▢
